Part 2 of 2

POST-SILICON VALIDATION Methodology In SoC

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Post-silicon validation is an essential step to verify the proper functioning and operation of an SoC, post manufacture. Part 2 discusses elaborately the various methods and parameters involving post-silicon validation

Post-silicon validation involves a number of activities including validation of both functional and timing behaviour as well as non-functional requirements. Each validation has methodologies to mitigate these.

Power-on debug. Powering on the device is actually a highly complex activity. If the device does not power on, the on-chip instrumentation architecture is typically not available, resulting in extremely limited (often zero) visibility into the design internals. This makes it difficult to diagnose the problem. Consequently, power-on debug includes a significant brainstorming component. Of course, some visibility and controllability exist even at this stage.

In particular, power-on debug typically proceeds with a custom debug board, which provides a higher configurability and fine-grained control over a large number of different design features.

The debug activity then entails coming up with a bare-bone system configuration (typically removing most of the complex features like power management, security and software/firmware boot mechanisms) that can reliably power on. Typically, starting from the time the silicon first arrives at the laboratory, obtaining a stable power-on recipe can take anywhere from a few days to a week.

Once this is achieved, the design is reconfigured incrementally to include different complex features. At this point, some of the internal design-for-debug (DfD) features are available to facilitate this process. Once the power-on process has been stabilised, a number of more complex validation and debug activities can be initiated.

Basic hardware logic validation. The focus of logic validation is to ensure that the hardware design works correctly and exercises specific features of constituent intellectual properties (IPs) in the system-on-chip (SoC) design. This is typically done by subjecting the silicon to a wide variety of tests including both focused tests for exercising specific features as well as random and constrained-random tests.

Traditionally, the SoC is placed on a custom platform (or board) designed specifically for debug, with specialised instrumentation for achieving additional observability and controllability of internals of different IPs. Significant debug software is also developed to facilitate this testing. In particular, tests executed for post-silicon validation are of system level, involving multiple IPs and their coordination.

Hardware/software compatibility validation. Compatibility validation refers to the activities that ensure the silicon works with various versions of systems, application software and peripherals. Validation accounts for various target use-cases of the system, platforms in which the SoC is targeted to be included and so on. Compatibility validation includes, among others, the following:

1. Validation of system usage with add-on hardware of multiple external devices and peripherals
2. Exercising various operating systems, applications, protocols and communication infrastructures

In addition to the generic complexities of post-silicon validation, a key challenge here is the large number of potential combinations (of configurations of hardware, software, peripheral, use-cases, etc) that need to be tested. It is common for compatibility validation to include over a dozen operating systems of different flavours, more than a hundred peripherals and over 500 applications.

Electrical validation. Electrical validation exercises electrical characteristics of the system, components and platforms to ensure an adequate electrical margin under worst-case operating conditions. Electrical characteristics include input-output, power delivery, clock and various analogue/mixed-signal (AMS) components. Validation is done with respect to various specification and platform requirements. For example, input-output validation uses platform quality and reliability targets.

As with compatibility validation, a key challenge is the size of the parameter space. For system quality and reliability targets, validation must cover the entire spectrum of operating conditions (voltage, current, resistance, etc) for millions of parts. The current